

1. If \$ P is invested at $r\%$ per year, compounded k times per year, the value after n years is

Sofia invests \$5000 in an account that pays 4% per year compounded annually.

Nickname:

2. Daniel invests \$12 000 in an account that pays 5.2% per year compounded **monthly**.

(a) Find the value of Daniel's investment at the end of 10 years.

Inflation is the rise in prices over time, so the same amount of money buys less in the future. If inflation is $i\%$ per year, then $\$A$ in n years' time buys the same as

$$\frac{A}{\left(1 + \frac{i}{100}\right)^n}$$

today. This is called the **real value** of $\$A$.

Over the 10 years, inflation is 2.8% per year.

(b) Find the real value of Daniel's investment at the end of 10 years.

(c) Hence find the real increase in the value of Daniel's investment.

(d) Daniel's sister keeps \$12 000 in cash at home for the same 10 years. Find the real value of her cash at the end of the 10 years.

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